

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Output:

```
geya@LenovoL480: ~/OS/Practical/Week 1$ lsblk
Mmio stale data:      Mitigation; Clear CPU buffers; SMT Host state unknown
Old microcode:        Not affected
Reg file data sampling: Not affected
Retbleed:              Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass:    Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:           Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:           Mitigation; IBRS; IBPB conditional; STIBP conditional; RSB filling; PBSRB-eIBRS Not affected
Srbds:                Unknown: Dependent on hypervisor status
Tsa:                  Not affected
Tsx async abort:      Mitigation; Clear CPU buffers; SMT Host state unknown
Vmscape:              Not affected

geya@LenovoL480:~/OS/Practical/Week 1$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M 1 disk 
sdb    8:16   0 159.4M 1 disk 
sdc    8:32   0    4G    disk [SWAP]
sdd    8:48   0    1T    disk /mnt/wslg/distro

geya@LenovoL480:~/OS/Practical/Week 1$ ps
PID TTY          TIME CMD
326 pts/0        00:00:00 bash
750 pts/0        00:00:00 ps

geya@LenovoL480:~/OS/Practical/Week 1$ top
top - 03:12:57 up 5 min, 1 user, load average: 0.06, 0.02, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 15931.6 total, 14780.7 free,  567.7 used,  781.1 buff/cache
MiB Swap: 4896.0 total,  4896.0 free,  0.0 used, 15363.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24284  15404 11636 S   0.0   0.1   0:00.82 systemd
    2 root        20   0   3180   2204  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0   3180   2048  1940 S   0.0   0.0   0:00.00 init
   47 root       19  -1  50396 16752 15460 S   0.0   0.1   0:00.22 systemd-journal
   82 systemd+   20   0  22540 14560 11936 S   0.0   0.1   0:00.10 systemd-resolve
   89 root       20   0  35252 12376  9236 S   0.0   0.1   0:00.32 systemd-udev
  146 root       20   0   2888   1948  1820 S   0.0   0.0   0:00.06 chronyd-starter
  147 root       20   0   4472   3152  2988 S   0.0   0.0   0:00.01 cron
  148 message+   20   0   8812   5412  4736 S   0.0   0.0   0:00.05 dbus-daemon
  155 root       20   0  44092 29840 14700 S   0.0   0.2   0:00.31 networkd-dispat
  160 root       20   0  18560   9540  8228 S   0.0   0.1   0:00.12 systemd-logind
  167 root       20   0 1792044 15152 12552 S   0.0   0.1   0:00.14 wsl-pro-service
```

```
geya@LenovoL480:~/OS/Practical/Week 1$ uname -a
Linux LenovoL480 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux

geya@LenovoL480:~/OS/Practical/Week 1$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 8
On-line CPU(s) list:   0-7
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz
CPU family:             6
Model:                 142
Thread(s) per core:    2
Core(s) per socket:    4
Socket(s):              1
Stepping:               10
BogoMIPS:               3791.99
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
e sse2 ss ht syscall nx pdpe1gb rdtscp lm arch_perfmon rep_good nopt xtopology cpuid tsc_kn
own_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 movbe popcnt aes xsave av
x f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp tpr_shadow ept
vpid ept_ad fsgsbase bmi1 hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap clflushopt xs
aveopt xsavec xgetbv1 xsaves vmmi md_clear flush_lld arch_capabilities

Virtualization features:
Virtualization:        VT-x
Hypervisor vendor:     Microsoft
Virtualization type:   full
Caches (sum of all):
  L1d:                  128 KiB (4 instances)
  L1i:                  128 KiB (4 instances)
  L2:                   1 MiB (4 instances)
  L3:                   6 MiB (1 instance)
NUMA:
NUMA node(s):          1
NUMA node0 CPU(s):    0-7
Vulnerabilities:
Gather data sampling:   Unknown: Dependent on hypervisor status
Ghostwrite:            Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit:         KVM: Mitigation: Split huge pages
L1tf:                  Mitigation; PTE Inversion; VMX conditional cache flushes, SMT vulnerable
Mds:                   Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown:              Mitigation; PTI
```

The Linux operating system acts as an interface between applications and hardware. Commands such as **lscpu**, **lsblk**, **ps** and **top** allow users to observe CPU, storage, and process information

without directly interacting with the hardware. The kernel manages these resources and provides abstractions such as processes, files, memory, and device interfaces, making hardware easier and safer for applications to use.